THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 4: **Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a quadratic equation? 1

(a) $x^2 + 3x + 2 = 0$ **(b)** $x^3 + 3x + 2 = 0$ **(c)** $3x + 2 = 0$ **(d)** $x + 3 = 2$

2. Is $x = 2$ a root of $x^2 - 5x + 6 = 0$? 1

(a) Cannot say **(b)** No **(c)** Only for $x > 0$ **(d)** Yes

3. In $2x^2 - 3x + 5 = 0$, the value of b is: 1

(a) -2 **(b)** -3 **(c)** -4 **(d)** -1

4. The roots of $(x - 2)(x - 3) = 0$ are: 1

(a) 2 and 3 **(b)** 2 and -3 **(c)** 5 and 6 **(d)** -2 and -3

5. The discriminant of $x^2 - 5x + 6 = 0$ is: 1

(a) 0 **(b)** 3 **(c)** 1 **(d)** 2

6. The nature of the roots of $x^2 - 5x + 6 = 0$ is: 1

(a) two distinct real roots **(b)** two equal real roots

(c) one real root and one imaginary root **(d)** no real roots

7. Which of the following is a quadratic equation? 1

(a) $x^2 - 5x + 6 = 0$ **(b)** $x^3 - 5x + 6 = 0$ **(c)** $-5x + 6 = 0$ **(d)** $x - 5 = 6$

8. Is $x = 4$ a root of $x^2 - 5x + 6 = 0$? 1

(a) Only for $x > 0$ **(b)** Cannot say **(c)** Yes **(d)** No

9. In $3x^2 + 7x - 2 = 0$, the value of b is: 1

(a) 6 **(b)** 8 **(c)** 9 **(d)** 7

10. The roots of $(x + 1)(x - 4) = 0$ are: 1

(a) -1 and 4 **(b)** 3 and -4 **(c)** -1 and -4 **(d)** 1 and -4

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Check whether $x = 2$ is a root of $x^2 - 7x + 10 = 0$. 2

12. Find the roots of $(x - 4)(x - 5) = 0$. 2

13. Find the discriminant of $x^2 - 6x + 8 = 0$ and state the nature of its roots. 2

14. Check whether $x = 3$ is a root of $x^2 - 7x + 10 = 0$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve by factorisation: $x^2 - 5x + 6 = 0$. **3**

16. Find the discriminant of $2x^2 - 4x + 3 = 0$ and hence the nature of its roots. **3**

17. The product of two consecutive positive integers is 20. Find the integers. **3**

OR

Solve by factorisation: $x^2 - 4x - 5 = 0$. **3**

18. Find the discriminant of $x^2 - 8x + 16 = 0$ and hence the nature of its roots. **3**

OR

The product of two consecutive positive integers is 42. Find the integers. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Solve by factorisation: $2x^2 - 7x + 3 = 0$. **5**

OR

The length of a rectangular plot exceeds its breadth by 5 m and its area is 66 m². Find its dimensions. **5**

20. Solve by factorisation: $3x^2 - x - 2 = 0$. **5**

OR

The length of a rectangular plot exceeds its breadth by 4 m and its area is 96 m². Find its dimensions. **5**

END OF QUESTION PAPER — Total: 40 Marks

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